

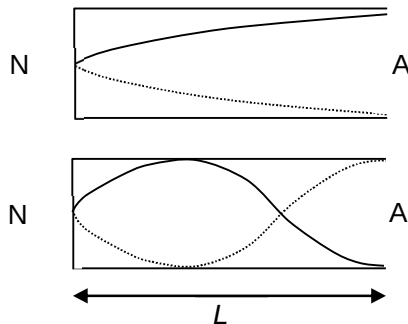
16 The two lowest modes of stationary waves set up are as shown:

$$f_1 = 300 \text{ Hz,}$$
$$\lambda_1 = v/f_1 = 4L$$

$$f_2 = ?$$
$$\lambda_2 = v/f_2 = (4/3)L$$

$$\lambda_1/\lambda_2 = f_2/f_1 = 3$$
$$f_2 = 3 f_1 = 900 \text{ Hz}$$

Answer: D



17 For double slit fringe separation x is given by